Literature review

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Role: the synthesized, full-length review prose for the methodological paper, organized by the argument in `docs/lit_review_plan.md` (§1–§7). It draws on the tagged `_Fiscal Shocks` Zotero library; the positioning argument it serves lives in `docs/brainstorm.qmd`, and the codebook-implementation distillations in `docs/codebook_sources.md`. As of the latest BetterBibTeX re-export every cited source now resolves to a `@`-citekey in `references.bib`; no `PENDING ACQUISITION` markers or plain-text "pending re-export" placeholders remain.

The methodological spine is four moves: take a hand-built narrative identification method (§2), scale it with LLMs (§4), validate the LLM as a measurement tool (§5), and hand off a generated variable for downstream inference (§6). Sections 2, 5, and 6 are the load-bearing pillars; sections 1, 3, and 7 motivate and situate.

Table 1: Coverage of the seven review sections in the current library.

Section	Role	Items	Status
§1 Motivation	Situate	13	Complete
§2 Narrative	Pillar	10	Strong
§3 Effects	Situate	12	Complete
§4 LLM text-as-data	Situate	17	Competitors + canon complete
§5 Validation	Pillar	14	Core gaps closed
§6 Generated-regressor	Pillar	4	De-risking trio filed
§7 Datasets	Situate	4	Complete

1. Motivation: why exogenous fiscal measures matter

The "why." For most emerging markets we lack consistent, comparable measures of exogenous fiscal shocks, the input credible fiscal analysis needs. Kept compressed; it does not lead the paper.

The motivating puzzle is fiscal procyclicality in the developing world: poor countries' fiscal policies tend to amplify rather than smooth their own crises. The stylized fact originates with Kaminsky et al. (2004) on "when it rains, it pours," and Vegh and Vuletin (2015) sharpen it on the tax side, documenting that tax *policy*, not just tax revenue, is procyclical in developing countries while acyclical in industrial ones. Crucially, they concede that the procyclical correlation "could just reflect the effect of tax multipliers" because their statutory-rate series carries no record of *motivation*, and they name the Christina D. Romer and Romer (2010) narrative classification as the missing input. This is the gap our pipeline fills.

The development-macro stakes are well established. Ilzetzki et al. (2013) show fiscal multipliers depend on country characteristics (exchange-rate regime, openness, debt), Frankel et al. (2013) document that institutional quality governs whether a country "graduates" from procyclicality, and Jha et al. (2014) find that in developing Asia tax cuts have larger countercyclical effects than spending, while conceding that roughly 70% of revenue variation is non-discretionary, the exact endogeneity that narrative motivation removes. The same team's work on fiscal space and the aftermath of financial crises (Christina D. Romer and Romer 2019a, 2017) underscores that the conduct of fiscal policy materially shapes downturn severity, raising the stakes for measuring it credibly. The broader identification stakes are framed by the credibility revolution (Angrist and Pischke 2010) and, in macro specifically, by the critique that macroeconomic identification remains opaque (P. Romer, n.d.).

What makes comparable cross-country shock series both scarce and valuable is the tax-composition–growth channel: Acosta Ormaechea et al. (2012) anchor the long-run link between the revenue mix and growth, and Crispolti et al. (2022) show, on a structured tax-reform database, that revenue yields vary systematically by instrument and by whether a reform changes rates or bases — exactly the instrument-level detail an exogenous-shock series must carry. Methodologically, the paper's architecture follows the Burke–Hsiang template of inventing one comparable cross-country unit and estimating a single transferable function over it (Burke et al. 2015; Hsiang et al. 2013).

2. Pillar: the narrative identification approach we are scaling

The method we automate. The methodological backbone, distinct from "effects" (§3) and "datasets" (§7).

The foundation is Christina D. Romer and Romer (2010) and its companion narrative analysis (Christina D. Romer and Romer, n.d.), which read historical documents to identify *why* taxes changed and to separate exogenous tax changes (motivated by long-run growth or deficit-reduction goals) from endogenous ones (responding to current or prospective output). The methodological doctrine, what makes narrative analysis rigorous, is stated most explicitly in Christina D. Romer and Romer (2023). These are distilled for codebook design in `docs/codebook_sources.md` §1.

The method travels. Cloyne (2013) applies the R&R strategy to the United Kingdom and finds "remarkably similar" magnitudes (a 1% tax cut raises GDP about 0.6% on impact and 2.5% over three years), the single most important precedent for our cross-country transfer claim. On the composition margin, Mertens and Ravn (2013) show personal income tax changes are far more potent for output than corporate ones, a decomposition only the narrative record enables.

The method is also contested, which is why §2 must engage scrutiny rather than only foundation. Mertens and Ravn (2014) prove that pure SVARs cannot recover the tax multiplier without assuming the output-elasticity of revenue, that narrative proxies used as instruments deliver larger multipliers (around 2 on impact, up to 3), and that measurement error in narrative series is itself consequential, a point that links directly to the generated-regressor problem in §6. Hebous and Zimmermann (2018) push the scrutiny further: the popular narrative tax measures are only weakly correlated with cyclically-adjusted revenues in the US and UK, and under weak-instrument-robust inference the implied multipliers are often insignificant, so the literature understates narrative-approach uncertainty. A parallel textual-analysis tradition in Germany (Latifi et al., n.d.; Tillmann et al. 2024) reads parliamentary speech rather than budget documents, a useful contrast to the document-based R&R lineage we follow. Narrative-inquiry methodology more broadly is surveyed by Kim (2015).

3. How these shocks get used: effects of fiscal policy

Shows the demand for the variable we produce, and motivates the §6 pillar (what happens when the regressor is LLM-measured).

The demand side is the multiplier literature. The SVAR tradition begins with Blanchard and Perotti (n.d.) and the narrative-vs-SVAR reconciliation runs through Mertens and Ravn (2014). State-dependence, multipliers that differ in recessions and expansions, is established by Auerbach and Gorodnichenko (2012), while Ramey and Zubairy (2018) supply the contrarian finding that spending multipliers stay below unity regardless of slack — so the regime-dependence is itself contested. The measurement pitfalls specific to tax multipliers are catalogued by Riera-Crichton et al. (2016), whose endorsement of R&R-style narrative identification directly underwrites the method we scale. The magnitude of multipliers across studies is synthesized by the meta-regression of Gechert (2015) and by Ramey's surveys (Ramey 2016, 2019). Ramey (2011) adds the anticipation/timing dimension that any fiscal-shock series must respect. On the tax side specifically, Dabla-Norris and Lima (2023) build a narrative dataset of 2,000+ consolidation-era tax measures and find base-broadening less contractionary than rate hikes — its yield/motivation/date coding closely parallels our own extraction targets.

The estimation toolkit our downstream payoff would use is local projections (Jordà 2005, n.d.), including the difference-in-differences event-study variant (Dube et al., n.d.). The authoritative gap citation closing this section is Ramey (2016): the fiscal portion of the handbook is essentially US-only, and the words "emerging" and "developing" do not appear.

4. LLMs as measurement tools: text-as-data foundations and deployment

Situates the tooling in the econ text-as-data canon and the social-science annotation literature, and introduces the direct competitors.

The econ text-as-data canon is anchored by Gentzkow et al. (2019), who also supply an adversarial flag we must answer: fixed text measures notoriously fail to transfer across domains, exactly the property our frozen, country-agnostic codebooks claim. Ash and Hansen (2023) survey the same lineage and foreground the challenge of *validating* algorithmic text output, the gap an H&K-style discipline closes, and Grimmer et al. (2022) give the book-length statement that measurement from text demands explicit validation rather than assumed validity. The meta layer for economists runs from Korinek (2025), who catalogues hallucination, error cascades, and prompt brittleness and pointedly declines to vouch for LLMs in causal inference, to the structured-extraction-and-scaling lens of Dell (2025); the NLP pipeline itself is reviewed for social scientists by Feuerriegel et al. (2025).

The social-science annotation literature supplies the precedent that LLMs can substitute for human coders. Gilardi et al. (2023) show zero-shot ChatGPT beats crowd workers on text annotation by about 25 points with higher intercoder agreement, and Törnberg (2025) that it beats both expert coders and supervised classifiers at multi-country political-affiliation labeling. Ziems et al. (2024) bound the claim, documenting taxonomic-labeling tasks where fine-tuned models still win and concluding LLMs augment rather than replace the pipeline. This is the validity our expert-agreement gates both lean on and stress-test in a higher-stakes, primary-document domain.

The direct competition is three LLM-based narrative fiscal papers that, read together, share one revealing shortcut. Das et al. (2026) map spending actions across 64 countries from EIU Country Reports but recover only qualitative direction, no %GDP magnitude, and defer taxes. Bhasin and Loungani (2026) extract austerity episodes from IMF Article IV reports for 17 OECD countries; they do attribute their LLM-versus-benchmark divergence to "shock sizing, not event selection," but frame it *oppositely* to a magnitude-failure reading — arguing their smaller, less-persistent sizes are better-identified than an inflated narrative benchmark — so our positioning should treat magnitude as the hard, contested step, not claim they concede a binding accuracy limit. Fritsch and Mazlish (n.d.) recover daily %GDP fiscal surprises with a tax-vs-spending split

(0.89 correlation with R&R tax shocks), but for the US only and from news wires building an *expectations* series. Each reads a secondary, harmonized, English-language source precisely to avoid the cost of primary national documents, and each validates by concordance against a pre-existing narrative series, a strategy that cannot travel to the countries the method is meant to serve. The (non-LLM) monetary analog Aruoba and Drechsel (n.d.) points to the alternative: validate against forecast-error predictability and theory-consistent impulse responses rather than an existing label set. Related LLM-as-measurement deployments outside fiscal policy include central-bank-communication analysis (Fernandez-Fuertes 2025) and the classification of research designs across 91,632 political-science articles (Torreblanca et al. 2025); the institutional uptake of LLMs across central-bank functions is surveyed by Araujo et al. (2024) and instantiated in domain-specialized models by Gambacorta et al. (2024).

5. Pillar: is the LLM output good enough to be a measurement?

The H&K spine plus the broader measurement-validity tradition, organized by our three sub-tasks: extraction, aggregation/entity-resolution, categorization.

The framework is Halterman and Keith (2025): a five-stage pipeline (prepare a codebook for humans and LLMs, test basic capabilities, evaluate zero-shot accuracy, analyze errors, and only then fine-tune) built on the premise that "codebook conceptualization is still a first-order concern." It is distilled for our behavioral tests and error analysis in `docs/codebook_sources.md` §3, and it supplies the central insight that off-the-shelf LLMs follow real codebooks imperfectly zero-shot.

The closest external validation template is Asirvatham et al. (2026) (the GABRIEL package), which separates two requirements our project conflates at its peril: **accuracy** (agreement with a benchmark) and **directness** (the measurement comes from the text, not from a correlated proxy). Their signal-stripping test, remove the rated content and confirm the rating collapses (a 98% reduction in their case), is the model for a high-value robustness check we should add to C2: mask the stated motivation and re-classify exogenous-versus-endogenous; if the call survives, the model is inferring from act name, era, or party rather than reading the rationale. They also reframe "is the model good enough" as human-parity (model-versus-consensus compared to leave-one-out human-versus-consensus), the correct frame for our no-ground-truth Phase 2 expert-agreement gates, and endorse the held-out discipline our S2/S3 split already enforces. Exemplars of LLM-as-measurement validated against expert judgment in other domains reinforce the pattern: multilingual psychological annotation (Rathje et al. 2024), which is directly relevant to our EN/BM stress test; physician-validated detection of stigmatizing clinical language (M. Singh 2026); and a traceable-reasoning diagnostic agent with 95.4% expert agreement (Zhao et al. 2026).

Two further sub-tasks of our pipeline draw on literatures beyond H&K. **Aggregation / entity resolution (C0)**, recognizing that "the Goods and Services Tax" and "Cukai Barang dan Perkhidmatan" are one act, is the classical record-linkage problem: anchored decision-theoretically by Fellegi and Sunter (1969), given its modern taxonomy by Binette and Steorts (2022), and shown to be reliably automatable in economics by Abramitzky et al. (2021), whose false-positive/match-rate frontier and human-versus-algorithm agreement directly parallel the over-merging-versus-fragmentation trade-off our canonical clustering manages. **Contamination / memorization**, the threat that historical US documents and the published R&R act list sit in pretraining and let a model recognize an act by name rather than reason from the supplied chunk, is quantified mechanistically by Carlini et al. (2022): extractable memorization rises with model size, training-data duplication, and prompt-context length, so the S1 memorization test must be re-run whenever the deployment model changes rather than treated as one-off. Detection techniques come from Golchin and Surdeanu (2024) and the surveying taxonomy from Deng et al. (2024). The **cross-lingual** risk our EN/BM and SEA legs face is documented by the multilingual-evaluation literature: Ahuja et al. (2023), H. Singh et al. (2024), and Xuan et al. (2025) all find marked degradation on lower-resource languages relative to English, and the latter two supply the parallel-item design (identical questions per language) that isolates the language effect from item difficulty in a like-for-like EN/BM comparison.

6. Pillar: using LLM-generated variables in downstream inference

The riskiest gap. Our output becomes a regressor in a VAR or local projection, making it a generated-regressor / measurement-error problem.

This is the pillar with the least coverage and the highest referee risk. The anchor is Ludwig et al. (2025), who frame precisely our case: an LLM-measured variable feeding a downstream causal estimate is an "estimation problem," and valid inference requires combining LLM outputs with a small labeled validation sample, not merely high classification accuracy. Three of their results bind on us, and are distilled in `docs/codebook_sources.md` §5:

- **Accuracy is not validity.** A model uniformly close to the truth can still produce arbitrarily biased downstream estimates; the bias equals the correlation of the LLM's error with the regressors. For us the dangerous regressor is the business cycle, the exact axis exogeneity hinges on.
- **Debias against a validation sample.** Do not plug raw LLM shocks into the multiplier regression; use the debiased estimator against a small random expert-labeled sample, which can yield tighter standard errors than the expert sample alone. Their simulations suggest 125–250 validation items suffice, a number infeasible at 20–40 Malaysian acts, forcing an honest design choice (fraction-of-corpus or pooled-regional validation, or a scope-limited inferential claim).
- **Specification search is a real risk.** Absent debiasing, the choice of model or prompt can change downstream coefficients in sign and significance, which directly implicates our Haiku-to-Sonnet C0 decision and any codebook iteration.

Three formal treatments of inference with a generated regressor now end this pillar's single-source exposure. Battaglia et al. (2024) is the most directly on point: they prove the two-step "estimate a latent variable from text, then regress on it" strategy is a measurement-error problem whose bias *re-centers* the estimator's asymptotic distribution, so ordinary (Eicker–Huber–White) standard errors have the right width but the wrong center and under-cover — and their explicit VAR/IRF extension is exactly our downstream object. The operative lesson is diagnostic: because the error mode is mis-centering, not mis-sizing, it cannot be caught by inspecting reported standard errors. Angelopoulos et al. (2023) (prediction-powered inference) and Egami et al. (2023) (design-based supervised learning) both convert a small gold-labeled subset plus a large LLM-labeled corpus into valid downstream inference *regardless* of model quality — PPI via a rectifier estimated on the gold sample, DSL via a doubly-robust pseudo-outcome over a *probability-sampled* gold subset. DSL's requirement that the validated subset be probability-sampled with a recorded sampling rate (not a convenience or borderline-acts sample) directly reshapes our expert-validation protocol and reframes the ≥80%/≥70% agreement targets as efficiency levers, not merely validity gates. Their shared limitation — little gain when predictions are weak or the unlabeled corpus is not large relative to the gold sample — is a live risk at 20–40 Malaysian acts, and the honest design constraint Phase 2 must confront. These are distilled for our protocol in `docs/codebook_sources.md` §5.

7. Existing fiscal datasets: the output landscape and comparators

What already exists, so our contribution and validation targets are legible.

The narrative / action-based lineage is the relevant comparator class. Its origin is the action-based fiscal-consolidation dataset of Pescatori et al. (2011), extended by Guajardo et al. (2014) (which shows action-based identification overturns the "expansionary austerity" reading that conventional cyclically-adjusted measures produce) and updated with %GDP sizing by Adler et al. (2024) — the OECD benchmarks that Das et al. (2026) and Bhasin and Loungani (2026) validate against. On the statutory / text-mined side, Crispolti et al. (2022) build on the IMF Tax Policy Reform Database and Vegh and Vuletin (2015) supply the statutory-rate panel. The US reference series we replicate and validate against is the Romer-Romer shock list, available as replication data (Christina D. Romer and Romer 2019b) and underlying `us_shocks.csv`.

Open sequencing note

§1 (motivation) and §3 (effects) are confirmed as motivation and demand framing, not application targets. Pillars §2, §5, and §6 carry the methodological contribution; with the generated-regressor trio (Battaglia, Angelopoulos, Egami) now filed and the §5 record-linkage, contamination, and cross-lingual cites resolved, all three pillars are multiply sourced and no `PENDING ACQUISITION` marker remains.

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